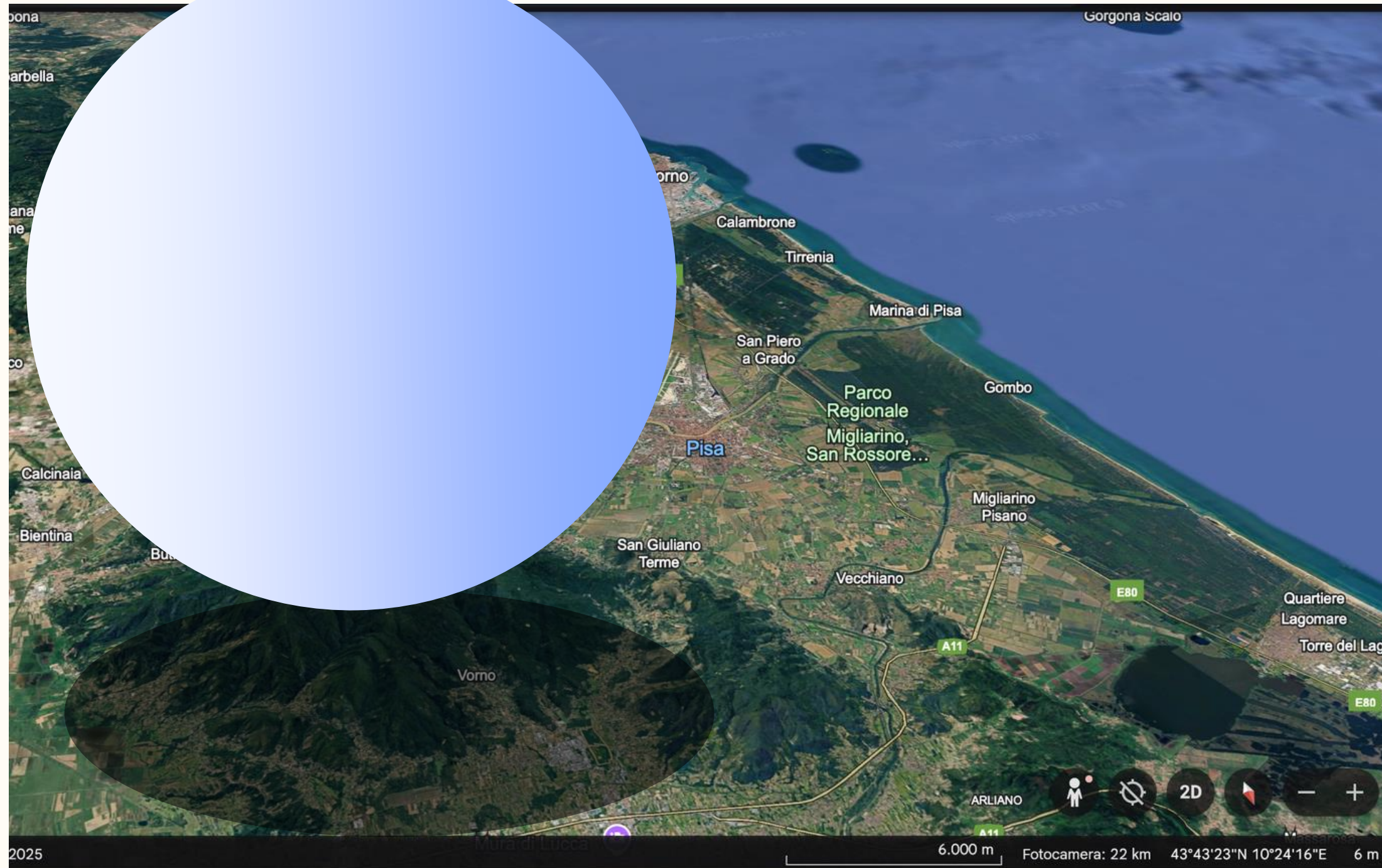




APPLICATIONS TO ASTROPHYSICS

Neutron stars



Some numbers

- ▶ Masses in the range $[1.4, 2.01] M_{\odot}$
- ▶ Radius between 10 km and 12 km

Extremely dense objects

COMPACT OBJECTS

- ▲ Internal density $\sim 10^{14} \text{ g/cm}^3$

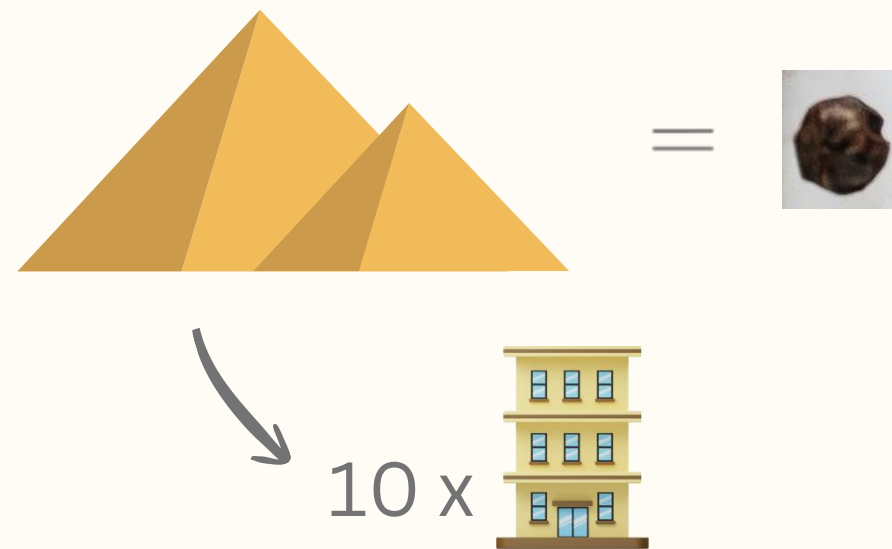
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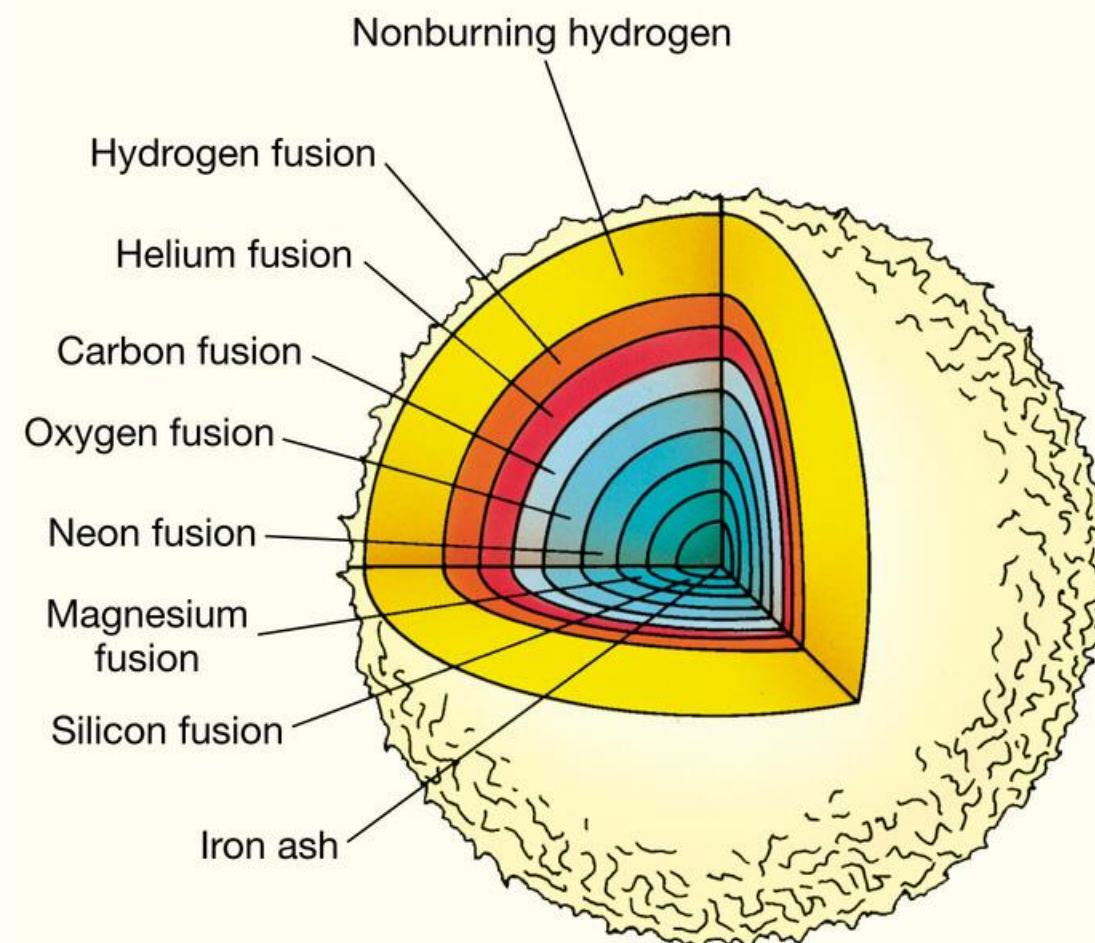


Same as if pyramid of Giza
squeezed down to a peppercorn



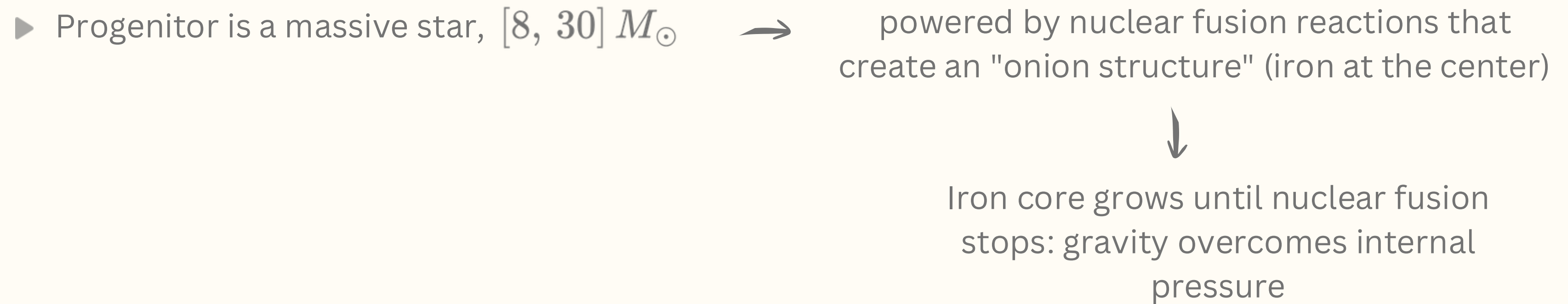
How do they form?

- Progenitor is a massive star, $[8, 30] M_{\odot}$ → powered by nuclear fusion reactions that create an "onion structure" (iron) at the center)

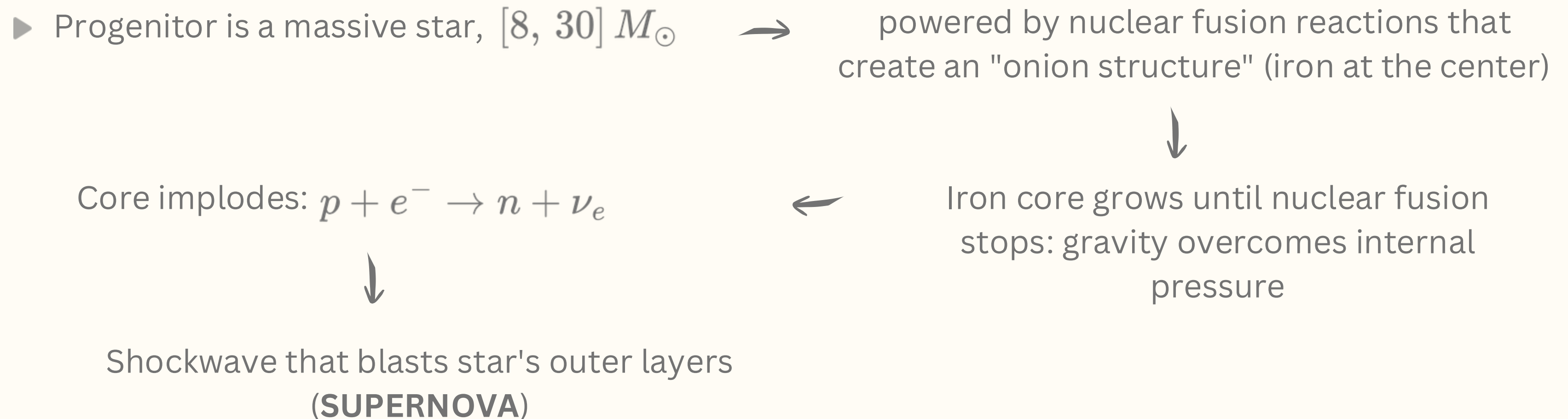


most stable nucleus, so can't
be fused further

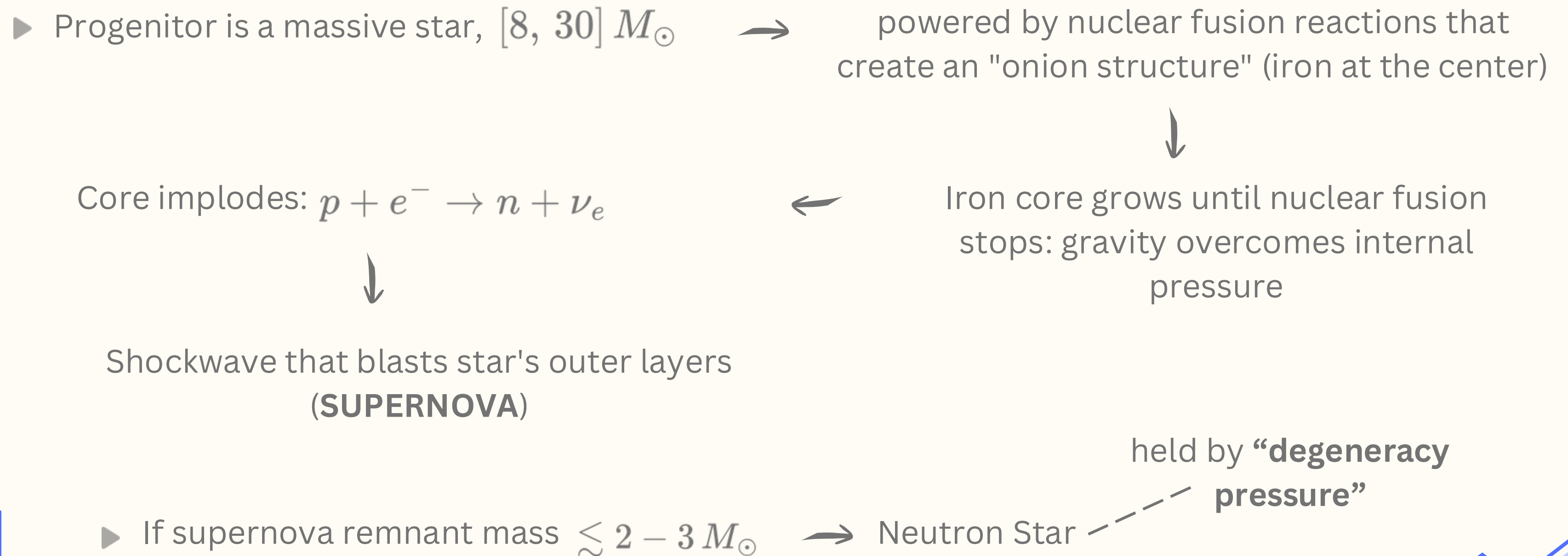
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APPLICATIONS TO ASTROPHYSICS

Neutron stars

How do they form?

- Progenitor is a massive star, $[8, 30] M_{\odot}$



powered by nuclear fusion reactions that create an "onion structure" (iron at the center)



Iron core grows until nuclear fusion stops: gravity overcomes internal pressure



Core implodes: $p + e^{-} \rightarrow n + \nu_e$



Shockwave that blasts star's outer layers
(**SUPERNOVA**)

Why this mass limit?

- If supernova remnant mass $\lesssim 2 - 3 M_{\odot}$



Neutron Star

held by "**degeneracy pressure**"

The TOV mass limit

$$2 - 3 M_{\odot}$$

► Tolman-Oppenheimer-Volkoff mass limit

► If Supernova remnant mass > TOV mass limit, degeneracy pressure can't prevent further gravitational collapse

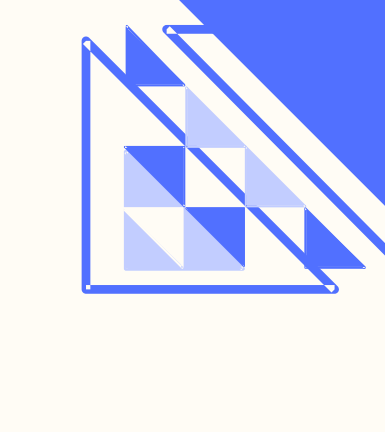


A Black Hole forms



APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”



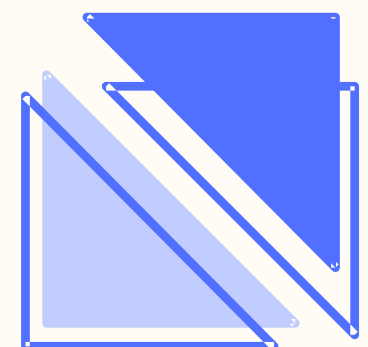
What are neutron stars made of?

- High density conditions in neutron stars cores:

Pauli exclusion principle forces
neutrons to occupy higher energy
states



Energetically cheaper for
neutrons to transform into
"hyperons"



APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”

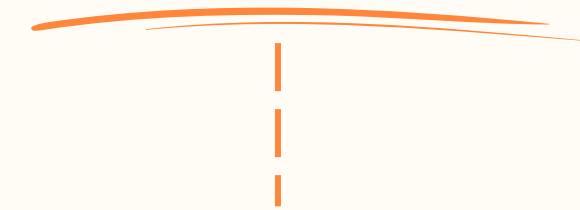
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like neutrons and
protons (**3 quarks**), but
at least one strange

!
lightest hyperon:

$$m_{\Lambda} = 1115.683 \pm 0.006 \text{ MeV}$$

APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”

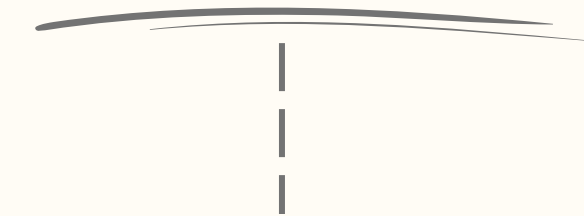
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Softening of the star equation of state

Hyperons appear and reduce internal pressure



lightest hyperon:

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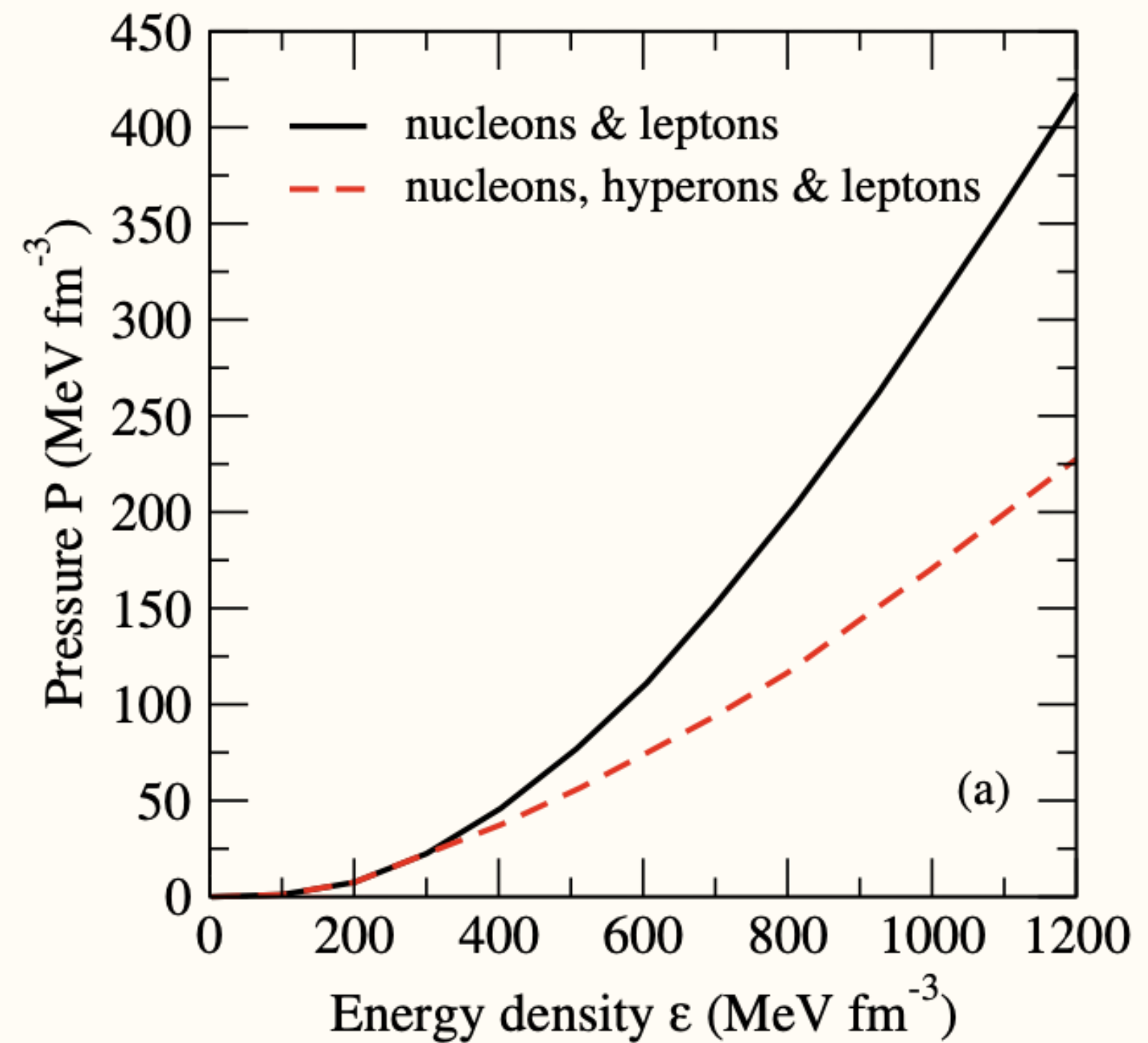
APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”

Softening of the star equation of state

EoS = relationship between pressure and energy density, inside the NS

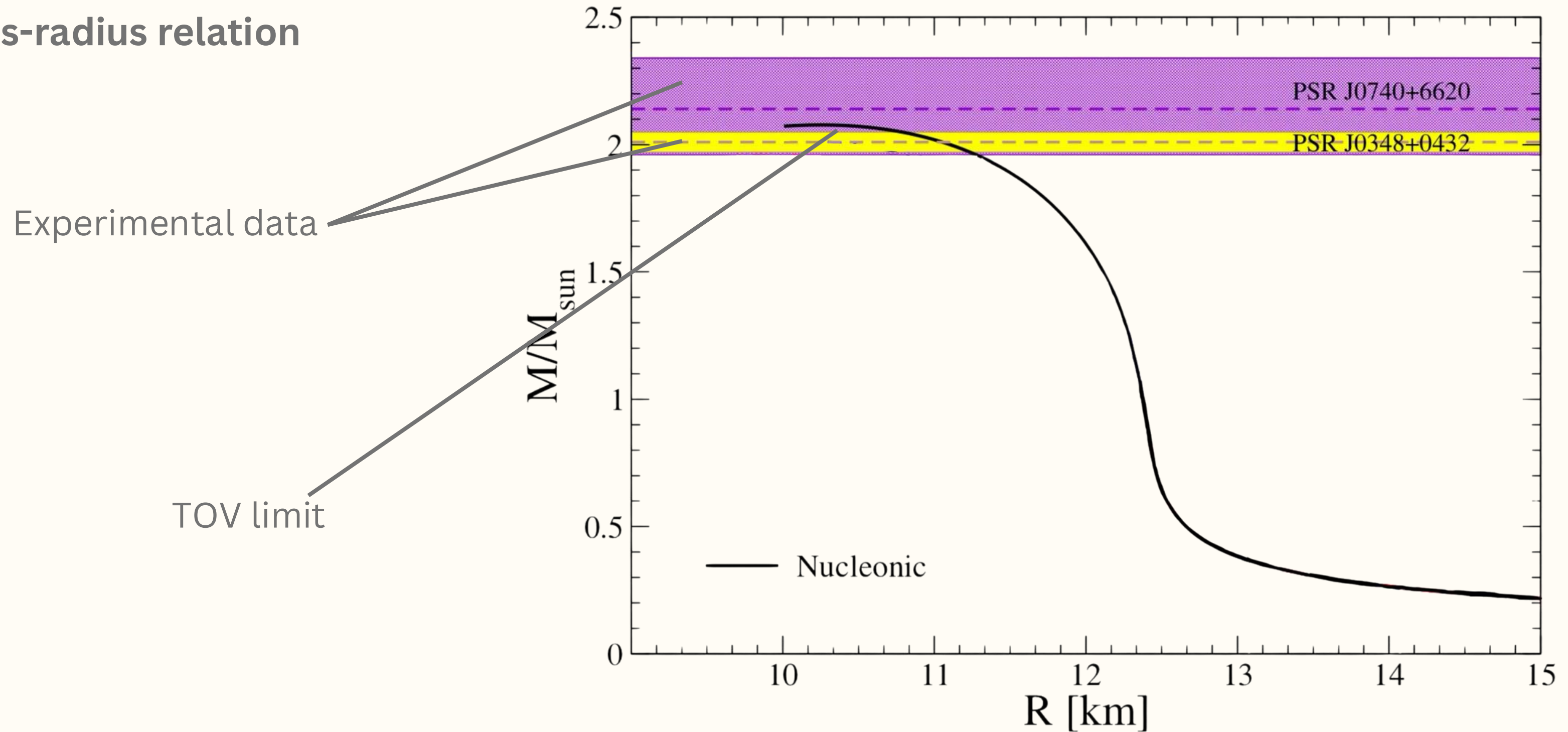
TOV mass limit depends on EoS model



APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”

Mass-radius relation



Adapted from D. Logoteta, I. Vidaña & I. Bombaci, *Eur. Phys. J. A* 55, 207 (2019)

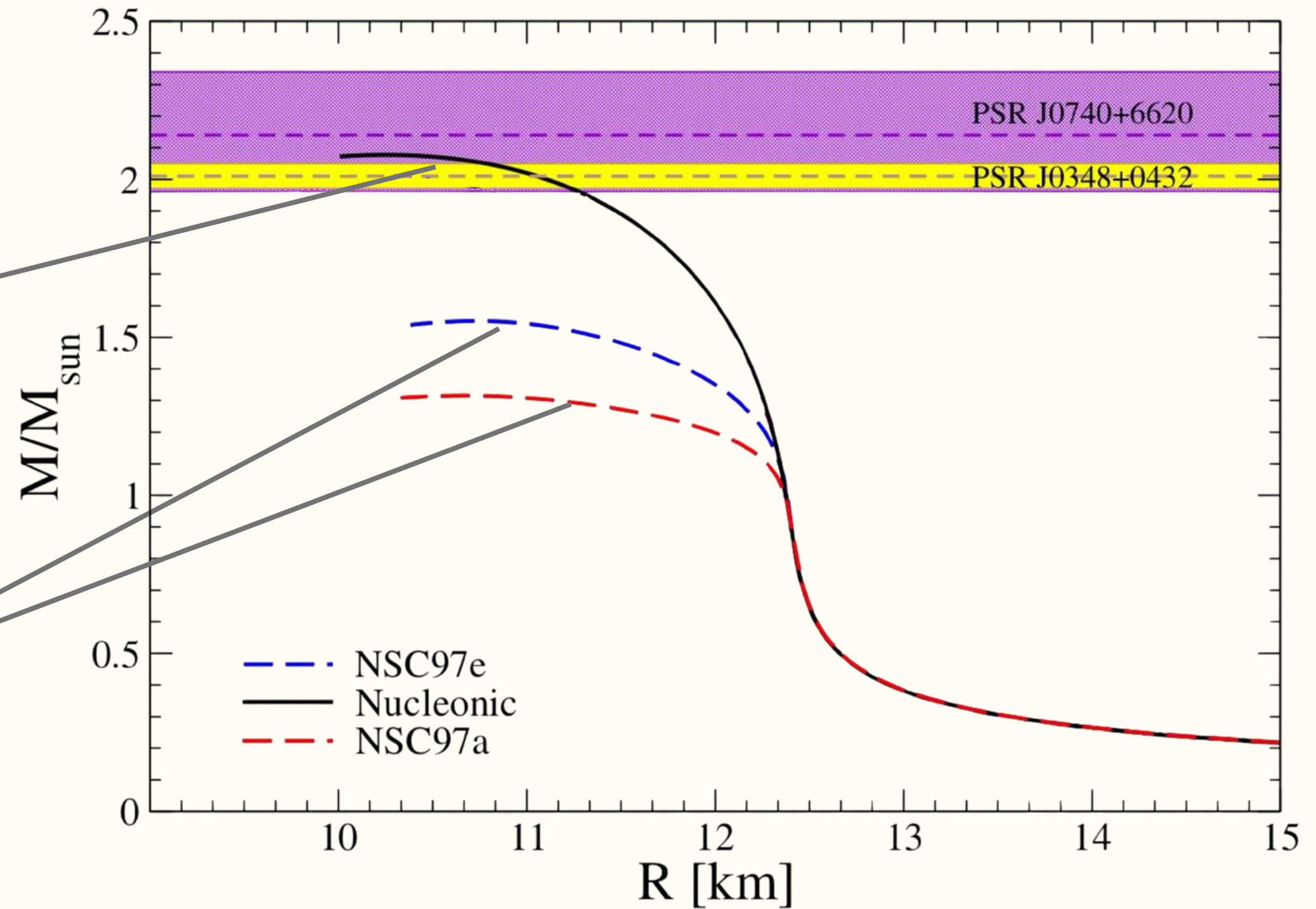
APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”

Mass-radius relation

Theoretical results
without hyperons

Theoretical results
with hyperons



Adapted from D. Logoteta, I. Vidaña & I. Bombaci, *Eur. Phys. J. A* 55, 207 (2019)

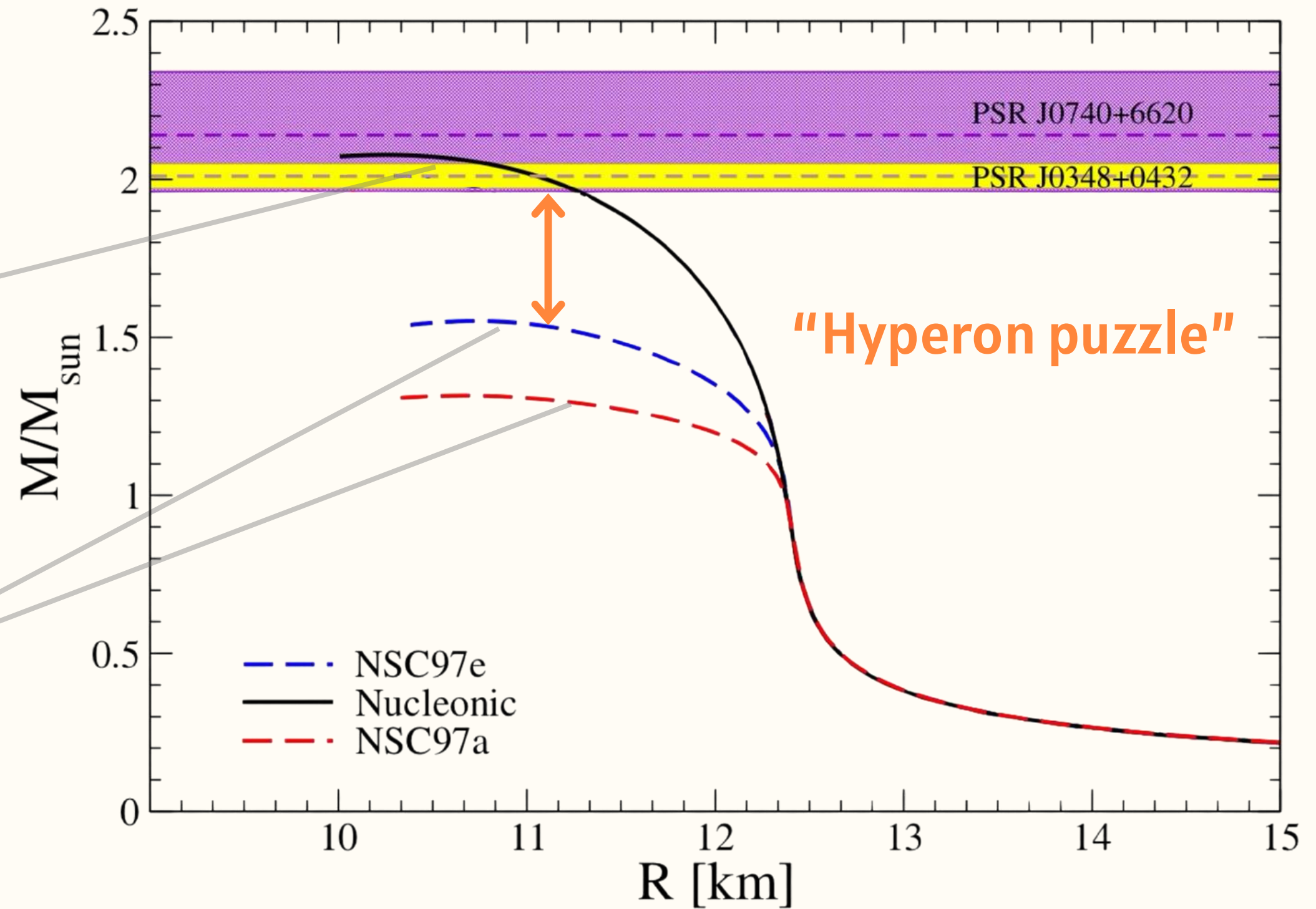
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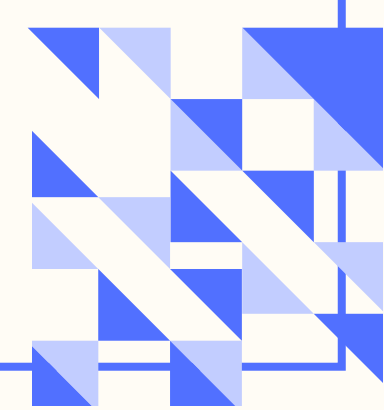
APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”

“Hyperon puzzle”

- ▶ Theory: hyperon formation should limit neutron stars to $\sim 1.5 M_{\odot}$
- ▶ Observations: exist neutron stars $\sim 2 M_{\odot}$

**How is nuclear
physics involved?**



APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”



“Hyperon puzzle”

- ▶ Theory: hyperon formation should limit neutron stars to $\sim 1.5 M_{\odot}$
- ▶ Observations: exist neutron stars $\sim 2 M_{\odot}$

How is nuclear
physics involved?

- ▶ Accurate description of the Neutron Star particle composition is needed

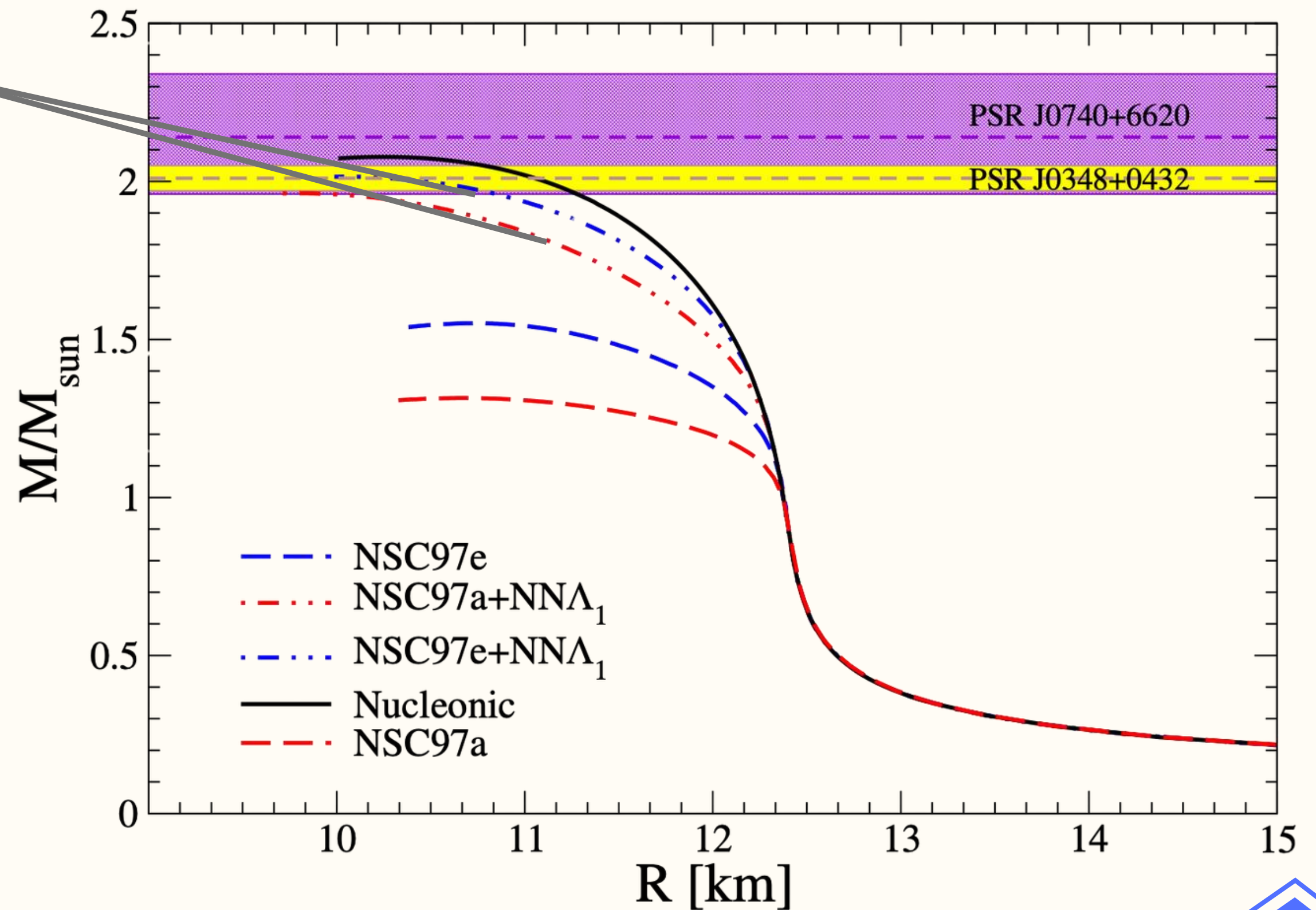


- ▲ Nucleon-nucleon (NN), hyperon-nucleon (YN) and hyperon-hyperon (YY) interaction
- ▲ 3-body forces (NNN, NNY, NYY)

APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”

Theoretical results with
hyperons + **3 body forces**



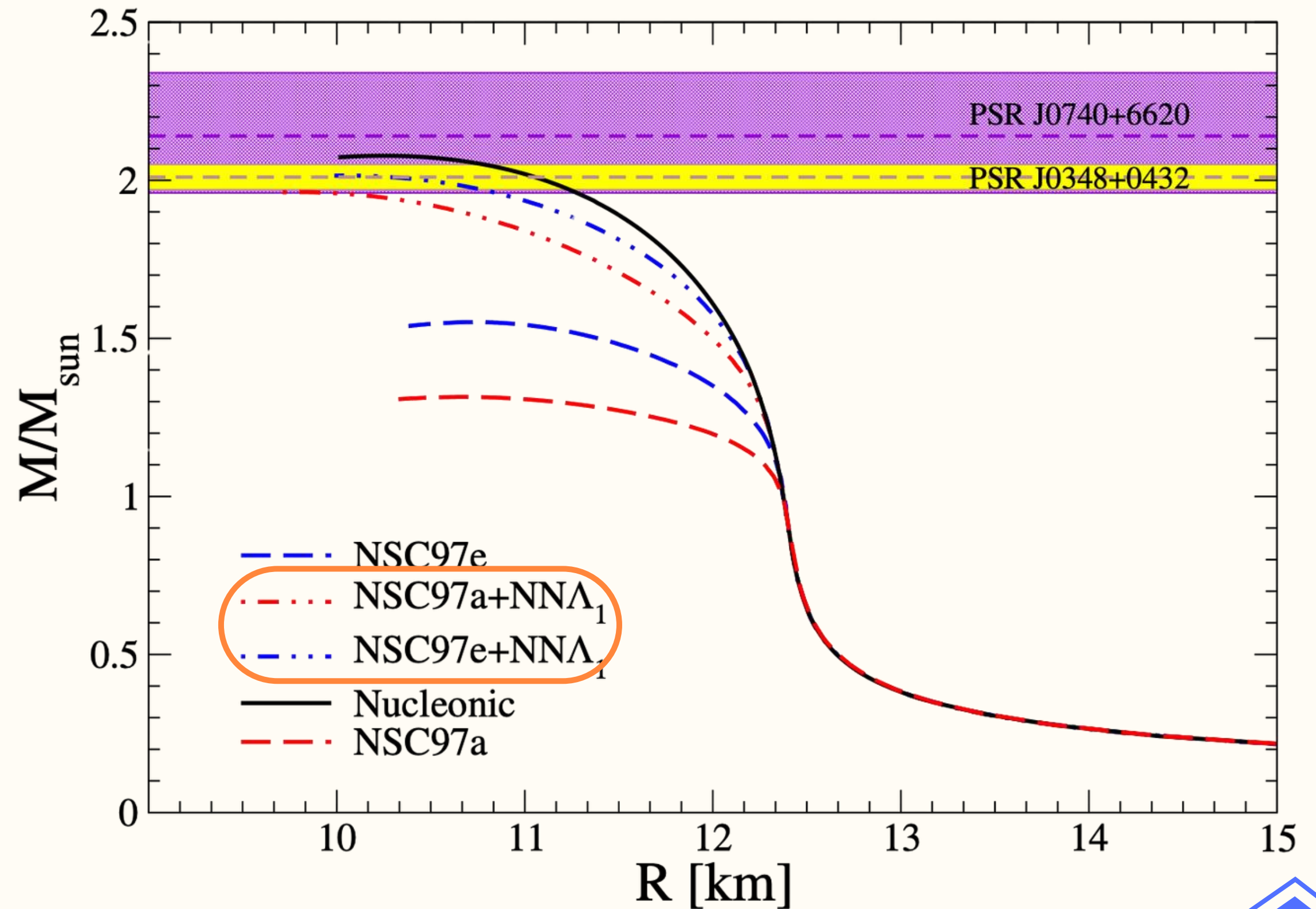
D. Logoteta, I. Vidaña & I. Bombaci, Eur. Phys. J. A 55, 207 (2019)

APPLICATIONS TO ASTROPHYSICS

The “hyperon puzzle”

are model dependent!

Theoretical results with
hyperons + 3 body forces



APPLICATIONS TO ASTROPHYSICS

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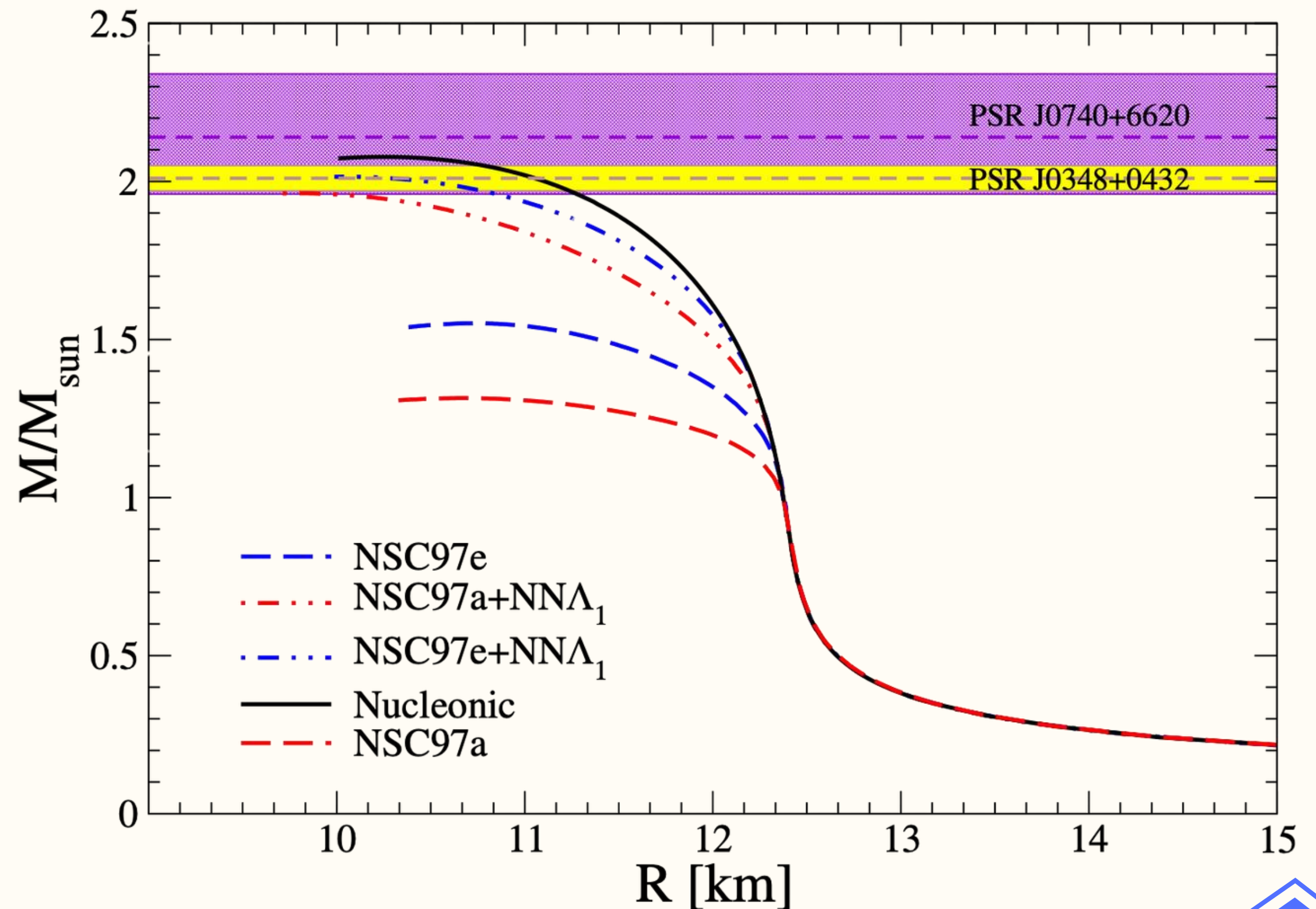
Theoretical results with
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Important to find a
solution for gravitational
wave physics applications



$\sim 10^4$ BNS
mergers/year with
new interferometers



D. Logoteta, I. Vidaña & I. Bombaci, Eur. Phys. J. A 55, 207 (2019)